DTU Physics
Department of Physics

Inverse Design of Light-Trapping Layers using Machine Learning

Master Thesis

Inverse Design of Light-Trapping Layers
using Machine Learning

Master Thesis
June 25, 2026

By
Andreas Nymand

Cover photo: Vibeke Hempler, 2012
Published by: DTU, Department of Physics, Building 307 & 309-313, 2800 Kgs. Lyngby, Fysikvej Denmark
https://physics.dtu.dk

Approval

This thesis has been prepared over six months at the Section for 2D-materials, 2DPHYS, Department of Physics, in collaboration with DTU Compute at the Technical University of Denmark, DTU, in partial fulfilment for the degree Master of Science in Engineering, MSc Eng.

It is assumed that the reader has a basic knowledge in the areas of physics and machine learning.

Andreas Nymand - S203903

....

Signature

....

Date

Abstract

This report details the surrogate optimization and active learning approaches used to inversely design light-trapping layers.

Acknowledgements

Andreas Nymand, MSc Eng, DTU
Author of this project

[Name], [Title], [affiliation]
[text]

[Name], [Title], [affiliation]
[text]

Contents

1 Introduction

This project explores the use of machine learning informed inverse design to beat classical bottlenecks in adjoint-based topology optimization for light-trapping layers in thin-film solar cells. In particular, I focus on replacing computationally expensive classical solvers with light-weight and scalable surrogate solvers, and I benchmark their performance, upsides and downsides. This chapter describes background on the topic, and motivates the use of surrogate solvers for inverse design of broadband and multimaterial

1.1 Thin-film solar cells and light-trapping layers

Where conventional solar cells typically are rigid cells made of $200\mu\mathrm{m}$ thick crystalline silicon or other highly absorbing materials in the visible light range, thin-film crystalline silicon solar cells are typically much thinner, in the range of 2 micrometres according to Machín and Márquez 2024. Crystalline silicon is a material of interest due to having ideal semiconductor properties for converting sunlight into electricity. Its material bandgap overlaps with a broad range of the solar spectrum, maximizing potential energy conversion efficiency, see Green 1982. But the absorbtion of light scales with the lights optical length in the solar cell. As such, thinner photovoltaic cells need ways to make light stay in the cell longer, to match the performance of thicker cells. According to an article by Amalathas and Alkaisi 2019, more than 90% of the global photovoltaic solar cell market is dominated by crystalline silicon based solar cells. The article also discusses how reductions in production cost and conversion efficiency improvements can be improved by light trapping approaches. Nanophotonic light trapping layers are structures on the scale of the wavelength of the light, that simultaneously seek to minimize reflections from and transmission through the solar cell. If the light can enter the material, but stay trapped within the active layer by total internal reflections or guided modes, the optical path length of the solar cell increases, and for the purpose of absorbing light, becomes thicker than it really is.

1.2 Topology optimization and the adjoint method

1.3 Fourier patterned light-trapping layers

2 Theory

The purpose of this chapter is to give a brief overview of the key theories that have informed my decision during the course of this project. Though the project has been mainly empirical through numerical studies and implementation, the theory provides invaluable insights for the design process and interpretation of results. First, I will discuss some theory of thin-film solar cells, leading to the design choices of my datset generation. First, I will discuss Fabry-Perot interferometers, as these share many optical properties with thin film-solar cells. I will then briefly introduce the theory behind rigorous coupled wave analysis (RCWA), which has been the workhorse of the generated data in this project. Since I have compared my results with the commercial finite-element method solver COMSOL, I will briefly introduce finite element method as well.

2.1 Thin-Film Solar Cells

2.2 Fabry-Perot Interferometers

2.3 Rigorous Coupled Wave Analysis

This subsection will introduce the concept of RCWA, a method in numerical photonics suited well for stacked periodic structures in the x and y axis with a period Λ_x, Λ_y respectively, and where each layer is homogeneous in the z-directionKim and Lee 2023. The structure's optical properties are completely described by the relative permitivity ϵ and relative permeability μ of each layer, meaning that each structure will have a geometry described by

$$\epsilon(x, y, z)_N = \epsilon(x + m \cdot \Lambda_x, y + n \cdot \Lambda_y)_N, m, n \in \mathbb{Z} \tag{2.1}$$

and

$$\mu(x, y, z)_N = \mu(x + m \cdot \Lambda_x, y + n \cdot \Lambda_y)_N, m, n \in \mathbb{Z}, \tag{2.2}$$

where m and n are integers and the subscript N denotes the N'th layer of the structure. For the entirety of this project, the designed fourier patterned light trapping layers will be designed using crystalline silicon (Si), Titanium Dioxide (TiO_2) and Silicon Nitride (Si_3N_4), which are all nonmagnetic, so the relative permeability $\mu = 1$. In general, ϵ and μ will be 3×3 tensors, but for isotropic and linear materials these are scalars. The equations solved for are of course the Maxwell equations, namely Faraday's law

$$\nabla \times \mathbf{E} = -\mu_0 \mu(x, y) \frac{\partial \mathbf{H}}{\partial t} \tag{2.3}$$

and Ampere's law

$$\nabla \times \mathbf{H} = \epsilon_0 \epsilon(x, y) \frac{\partial \mathbf{E}}{\partial t}, \tag{2.4}$$

Where ϵ_0 and μ_0 are the vacuum permitivity and vacuum permeability respectively. The structures are assumed free of charge and current,

$$\nabla \cdot (\epsilon \mathbf{E}) = 0$$

and

$$\nabla \cdot (\mu \mathbf{H}) = 0.$$

At a single frequency and in homogeneous media, the eigenmodes of Maxwell's equations are solved by plane waves, see Griffiths 2017 section 9.2.1 for example, meaning

$$\mathbf{E} = \mathbf{E}_0 \exp i(\mathbf{k} \cdot \mathbf{r} - \omega t) \tag{2.5}$$

and

$$\mathbf{H} = \mathbf{H}_0 \exp i(\mathbf{k}\cdot\mathbf{r} - \omega t), \tag{2.6}$$

Where $\mathbf{E}_0$ and $\mathbf{H}_0$ are the (complex) amplitudes of the fields, with the physical fields being the real parts of $\mathbf{E}$ and $\mathbf{H}$, $\mathbf{k}$ being the wavevector, i.e. the spatial frequency, and ω being the frequency of the fields, also given by the dispersion relation $\omega = \frac{c|\mathbf{k}|}{n(\lambda)}$, where λ is the wavelength of the light. It is noted that conservation of momentum leads to the spatial components of the wavevector only having two dgrees of freedom, i.e.

$$|\mathbf{k}| = \frac{2\pi n(\lambda)}{\lambda} = \sqrt{k_x^2 + k_y^2 + k_z^2} = k_0, \tag{2.7}$$

and in general, for an incident angle θ and azimuthal angle ϕ, the x and y components will be chosen by

$$(k_x, k_y) = k_0(\sin\theta\cos\phi, \sin\theta\sin\phi). \tag{2.8}$$

Inserting equations 2.5 and 2.6 in equations 2.3 and 2.4 converts the time derivative to an algebraic factor of $-i\omega t$, meaning that the equations appear in their time harmonic form, common to many frequency domain numerical methods in photonics,

$$\nabla \times \mathbf{E} = i\omega\mu_0\mu(x, y)\mathbf{H} \tag{2.9}$$

$$\nabla \times \mathbf{H} = -i\omega\epsilon_0\epsilon(x, y)\mathbf{E}. \tag{2.10}$$

Owing to the periodicity of the system, see equations 2.1 and 2.2, both the electromagnetic fields and the permitivity and permeability may be written as their truncated fourier series in the x- and y-directions as below. Truncating the electromagnetic fields to order M, N in x, y makes RCWA a semi-analytical method precise only to this fourier order. The final fields are reconstructed as an inverse Fast Fourier Transform, so including enough fourier orders is an important convergence criterion.

$$\epsilon(x, y) = \sum_{m=-2M}^{2M} \sum_{n=-2N}^{2N} \epsilon_{m,n} \exp i(mL_x x + nL_y y), \tag{2.11}$$

$$\mu(x, y) = \sum_{m=-2M}^{2M} \sum_{n=-2N}^{2N} \mu_{m,n} \exp i(mL_x x + nL_y y), \tag{2.12}$$

$$\mathbf{E} = \exp i\mathbf{k}\cdot\mathbf{r} \sum_{m=-M}^{M} \sum_{n=-N}^{N} \mathbf{E}_{m,n} \exp i(mL_x x + nL_y y), \tag{2.13}$$

$$\mathbf{H} = \exp i\mathbf{k}\cdot\mathbf{r} \sum_{m=-M}^{M} \sum_{n=-N}^{N} \mathbf{H}_{m,n} \exp i(mL_x x + nL_y y), \tag{2.14}$$

where $L_x = \frac{2\pi}{\Lambda_x}$ and $L_y = \frac{2\pi}{\Lambda_y}$, and the parameters $\epsilon_{m,n}, \mu_{m,n}, \mathbf{E}_{m,n}$ and $\mathbf{H}_{m,n}$ are the fourier amplitudes of the m, n'th diffraction order. It is worth mentioning, that since the momentum of light is always conserved, k_z implicitly depends on the expansion orders m, n, and writing out the k-vector in full, including the fourier components, yields

$$\mathbf{k}_{m,n} = (k_x + mL_x, k_y + mL_y, k_z(m, n)) = \tag{2.15}$$

$$= \left(k_x + mL_x, k_y + mL_y, \sqrt{k_0^2 - (k_x + mL_x)^2 - (k_y + mL_y)^2}\right), \tag{2.16}$$

where equation 2.7 has been used. The physical interpretation of the fourier series of the electromagnetic fields is then made obvious; the different terms are different diffraction orders, each with part of the initial z-component momentum kicked into either the x- or y-direction. It is noted how the permitivity and permeability is expanded to twice the order of the electromagnetic fields E and H. This arises from the fact that the algebraic components of equations 2.3 and 2.4 include products of Fourier series, so to reach order M, N in the electromagnetic fields, the permitivity and permeability must be truncated at twice the order. An example for just one fourier order will be given below, but for more details, I refer to Hwi Kim 2012. By inserting

2.4 Finite Element Method

3 Methods

3.1 Numerical implementation

3.2 Convergence tests

3.3 Machine Learning Surrogates

3.4 Optimization Strategies

4 Results

Present the performance of the surrogate models and the final inverse design outcomes.

Figure 4.1: Placeholder for a results figure.

5 Discussion

Discuss the results and what worked/didn't work.

6 Outlook

Discuss future work.

7 Conclusion

Summarize the findings.

Bibliography

Machín, A. and F. Márquez (Mar. 2024). "Advancements in Photovoltaic Cell Materials: Silicon, Organic, and Perovskite Solar Cells". In: *Materials (Basel)* 17.5. PMID: 38473635; PMCID: PMC10934213, p. 1165. DOI: 10.3390/ma17051165.

Green, M A (Jan. 1982). *Solar cells: operating principles, technology, and system applications*. Englewood Cliffs, NJ, United States: Prentice-Hall, Inc. URL: https://www.osti.gov/biblio/6051511.

Amalathas, Peter and Maan M. Alkaisi (Sept. 2019). "Nanostructures for Light Trapping in Thin Film Solar Cells". In: *Micromachines* 10.9, p. 619. DOI: 10.3390/mi10090619.

Kim, Changhyun and Byoungho Lee (2023). "TORCWA: GPU-accelerated Fourier modal method and gradient-based optimization for metasurface design". In: *Computer Physics Communications* 282, p. 108552. ISSN: 0010-4655. DOI: https://doi.org/10.1016/j.cpc.2022.108552. URL: https://www.sciencedirect.com/science/article/pii/S0010465522002715.

Griffiths, David J. (2017). *Introduction to Electrodynamics, fourth edition*. Cambridge University Press.

Hwi Kim, Junghyun Park Byoungho Lee (2012). *Fourier Modal Method and Its Applications in Computational Nanophotonics*. CRC press.

A Appendix

Technical
University of
Denmark

Building 307 & 309-313
2800 Kgs. Lyngby, Fysikvej
Tlf. 4525 1700

https://physics.dtu.dk